PRODUCT REQUIREMENTS DOCUMENT (PRD) & SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project: NutriNext - "One Kitchen, Many Plates"

Target Event: Bharat Innovation Challenge (BIC) 2025 - Grand Finale **Document Version:** 1.0.0 (Enterprise Final) **Classification:** Confidential / Technical Blueprint

1. EXECUTIVE SUMMARY & PRODUCT VISION

1.1 The Core Problem Current digital health interventions operate on an isolated, individualistic paradigm. They fail in the Indian socio-cultural context where households utilize a single kitchen to cater to multi-generational demographics with conflicting health parameters (e.g., pediatric growth, geriatric diabetes, maternal anemia).

1.2 The NutriNext Solution NutriNext is an AI-powered Health Operating System (Health OS). It leverages a dual-context architecture—a permanent Static Profile intersecting with a weekly Dynamic Generation Popup—to feed a localized "Harmony Algorithm." This engine outputs a single base family meal with micro-adjusted serving instructions per member, enforcing compliance with the ICMR-NIN 2024 Dietary Guidelines, optimizing quick-commerce grocery budgets, and eliminating the psychological burden on the primary caregiver.

2. SYSTEM ARCHITECTURE & TECH STACK

2.1 Infrastructure & Compliance

- **Authentication & Database:** Supabase (PostgreSQL).
  - *Critical Compliance Update:* Hosted strictly in the **AWS Mumbai (ap-south-1) Region** to ensure absolute compliance with the Indian Digital Personal Data Protection (DPDP) Act, 2023. No patient health data (PHI) leaves the Indian sovereign borders.
- **Frontend:** React Native (Cross-platform iOS/Android) for high-performance edge rendering.

2.2 Artificial Intelligence & Machine Learning Layers

- **Core Logic Engine:** Google Vertex AI Search for Healthcare & Gemini 2.0 Flash (Ensures HIPAA-grade security, minimizes hallucination, enables high-speed multimodal reasoning).
- **Vernacular Voice Processing:** Sarvam AI APIs (Translates and transcribes regional languages like Hindi into structured English JSON for the LLM).
- **Edge Vision Processing:** EfficientNet-B4 (Lightweight computer vision model deployed on-device for leftover identification, integrated with a Semantic Error Propagation Control

mechanism).

3. APP SECTION 1: PROFILE CURATION (THE STATIC ENGINE)

Module Goal: Establish the permanent socio-cultural and biological baseline of the household.

3.1 Onboarding Modalities Users can bypass traditional UI friction via two methods:

1. **Manual Form Entry:** Standard UI dropdowns and toggles.
2. **Multilingual Voice Agent:** User speaks naturally (e.g., *"Hum 4 log hain, papa ko sugar hai"*). Sarvam AI transcribes and translates, while Gemini parses the intent to auto-fill the database matrix.

3.2 Data Schema: Family Level (`req_fam_profile`)

- `family_count` (Integer): Total members.
- `region` (String): State/Region (influences seasonal availability and staple grains).
- `lang_pref` (Enum): Hindi / English.
- `dietary_baseline` (Enum): Strictly Veg / Veg + Eggs / Non-Veg / Mixed.
- `meal_frequency` (Enum): 2 meals / 3 meals / 3 meals + snacks.
- `cooking_skill` (Enum): Beginner / Intermediate / Experienced.

3.3 Data Schema: Member Level (`req_mem_profile`)

- `member_name` (String) | `age` (Integer) | `gender` (Enum)
- `height_cm` (Float - Optional) | `weight_kg` (Float - Optional)
- `activity_level` (Enum): Sedentary / Light / Moderate / Active
- `primary_goal` (Enum): Weight Loss / Weight Gain / Maintain / Build Muscle / Manage Condition / None.
- `goal_pace` (Enum): Slow (0.25kg/wk) / Moderate (0.5kg/wk). *Logic: Only render if goal = Loss/Gain.*
- `health_conditions` (Array, Max 2): Diabetes Type 2 / Hypertension / Anaemia / Obesity / High Cholesterol / Hypothyroid / PCOS / Kidney Issues.
- `dietary_type` (Enum): Strictly Veg / Jain / Eggetarian / Non-Veg / Occasional Non-Veg.
  - *Sub-Logic (If Occasional):* `allowed_days` (Array: Mon-Sun) + `allowed_proteins` (Array: Chicken, Mutton, Fish, Eggs).
- `cultural_rules` (Enum): No Beef / No Pork / Sattvic (No Onion/Garlic) / Jain.
- `spice_tolerance` (Enum): Mild / Medium / Spicy.
- `allergies` (Array): Peanuts / Dairy / Gluten / Tree Nuts / Shellfish / Soy / Sesame.

- `dislikes` (String Array, Max 5): Free text (e.g., "Karela, Brinjal").
- `fasting_baseline` (Array): Weekly days (Mon-Sun) + `ekadashi_observance` (Boolean).
- `festival_alerts` (Boolean): App syncs with local panchang/calendar to suggest fasting menus.
- `tiffin_required` (Enum): No / Yes (School) / Yes (Office).

3.4 Quantiphi Responsible AI (RAI) Auto-Assignment Rules *The system must strictly execute these logical overrides to prevent algorithmic harm:*

- **Rule A (Pediatric):** `IF age < 5 THEN` hide goal field, auto-assign `goal = early_childhood_nutrition` .
- **Rule B (Growth):** `IF age >= 5 AND age <= 12 THEN` hide goal field, auto-assign `goal = healthy_growth` .
- **Rule C (Teen Anti-Dysmorphia):** `IF age >= 13 AND age <= 17 THEN` remove "Weight Loss" from array. Restrict to Maintain/Build Muscle/Manage Condition.
- **Rule D (Geriatric):** `IF age >= 60 THEN` auto-assign `goal = senior_nutrition` (can be user-overridden).
- **Rule E (ICMR Baselines):** `IF weight != NULL AND goal != NULL THEN` trigger ICMR-NIN basal metabolic calculation script to display daily caloric target to user.

4. APP SECTION 2: MEAL GENERATION (THE DYNAMIC MATRIX)

Module Goal: Capture the localized, weekly context to override static defaults.

4.1 The Weekly Context Popup (`req_weekly_context` **)** Triggered prior to generating the weekly plan. Pre-filled with static profile data; user modifies only deviations.

- **Financial Context:** * `budget_in_inr` (Integer): Monthly grocery budget constraint.
  - `dining_out_freq` (Enum): None / 1-2 times / Frequently. (Algorithm deducts meals accordingly).
- **Temporal Context:**
  - `weekday_prep_time` (Enum): <20 mins / 20-40 mins / No limit.
  - `weekend_prep_time` (Enum): Quick / Elaborate / No preference.
- **Member Contextual Overrides (Iterated per member):**
  - `weight_update` (Float): "Any change since last time?" (Only if goal is active).
  - `health_status_this_week` (String): Free text (e.g., "Heavy cold", "Bloating"). Passed as a high-priority system instruction to Gemini to alter spice/temperature vectors.
  - `fasting_this_week` (Array): User override for specific upcoming fasts.

    - `tiffin_this_week` (Boolean): Override for holidays/sick days.

- **The Global Override:**
    - `special_instruction` (Text): e.g., "More protein for Papa this week," "Use up the spinach." This acts as the absolute highest-weighted parameter in the Gemini API prompt.

5. APP SECTION 3: PANTRY SCAN & INVENTORY MANAGEMENT

Module Goal: Frictionless logging of available ingredients to drive zero-waste algorithms.

5.1 Trimodal Logging System

1. **Voice (Sarvam AI):** User speaks: "Mere paas aadha kilo tamatar aur kal ka chawal hai."
2. **Chat (Text):** Standard chat interface.
3. **Computer Vision (The Unfair Advantage):** * User uploads a photo of the open fridge or leftovers.
    - **Semantic Error Propagation Control:** The EfficientNet-B4 edge model scans the image. If `confidence_score < 0.92` (e.g., distinguishing white salt from rice flour), the AI halts generation and triggers a Sarvam AI voice prompt: *"I see leftover Dal, but am unsure about the white powder. Is it salt or flour?"*
    - Once confirmed, the system calculates exact leftover volume.

5.2 Dynamic Budget-Optimization Engine (Swiggy Hook)

- The system cross-references the required ingredient list against a simulated Quick-Commerce API (fetching prices from local Mandi databases like Agmarknet, marked up to retail).
- **Algorithmic Arbitrage:** If local tomato prices spike in Bokaro from ₹40 to ₹80, the system autonomously rewrites Thursday's recipe, swapping tomatoes for tamarind/pumpkin puree to maintain acidity and budget, alerting the user to the savings.

6. CORE ALGORITHM: THE HARMONY ENGINE (ICMR-NIN 2024 GROUNDING)

Module Goal: Generate one base meal and modify it across multiple plates using Indian clinical standards.

6.1 System Prompt Architecture (Sent to Gemini 2.0) The LLM receives a synthesized JSON containing: `[Static Profile] + [Weekly Context] + [Pantry Inventory] + [Budget Constraint]`.

6.2 Hardcoded ICMR-NIN 2024 Guardrails The LLM output MUST pass through these mathematical validation gates before rendering to the user:

- **Bioavailable Amino Acid Complementarity:** The base meal algorithm must ensure that cereal and pulse raw weights are formulated in an exact **3:1 ratio** to naturally maximize lysine and methionine absorption, minimizing the need for commercial protein powders (ICMR 2024 warning).
- **NOVA/HFSS Thresholds:** No generated recipe (or logged cheat meal) can exceed 250 kcal/100g, or surpass 5g of sodium per day across the family plan.
- **Plate Micro-Adjustments:** * *Scenario:* Family makes Dal Khichdi.
  - *Mother (Anaemia):* Algorithm multiplies her portion of Green Leafy Vegetables (GLV) to hit the newly mandated 100g/day ICMR target.
  - *Father (Diabetes):* Algorithm instructs the cook to replace 50% of his specific rice portion with foxtail millet (low glycemic index).

7. APP SECTION 4: RECIPES & BIOCHEMICAL PRE-PROCESSING

Module Goal: Guide the user through the cooking process with scientific precision.

7.1 Standard Output

- Step-by-step instructions.
- Portion multipliers (e.g., "Serve Papa 1 katori, serve Rohan 2 katoris with 1 tsp extra ghee").

7.2 Biochemical Pre-Processing Module (BPM)

- Indian vegetarian diets contain anti-nutritional factors (phytates, oxalates) that block iron/calcium absorption.
- NutriNext features an **Asynchronous Time-Shifted Alert System**.
- *Logic:* If the Tuesday lunch algorithm schedules Rajma (kidney beans), the app triggers a push notification on Monday at 8:00 PM: *"Reminder: Soak the Rajma now. Overnight soaking reduces phytates by 40%, unlocking iron for Mom's anemia goals."*

8. APP SECTION 5: AI-CHAT & BEHAVIORAL LOGGING

Module Goal: Handle dietary deviations dynamically without punitive measures.

8.1 The "Cheat Meal" Adjuster

- User logs an out-of-plan event via chat/voice: *"Ate 2 samosas at a wedding today."*
- The LLM calculates the caloric and sodium load (using ICMR databases).
- **Rolling Adjustment:** The system silently recalculates the next 48 hours of meals, suppressing sodium and simple carbs, without sending negative/punitive alerts.

9. APP SECTION 6: HEALTH, IoT, & GAMIFICATION

Module Goal: Long-term retention via behavioral economics and hardware integration.

9.1 Hardware Integration

- Bluetooth API connections to smart weighing scales and Continuous Glucose Monitors (CGMs).

9.2 Libertarian Paternalism & The Family Harmony Score

- Diets fail due to the "caregiver burden" and the isolation of sick family members.
- NutriNext tracks a **Family Harmony Score**.
- Points are awarded universally when:
  - The teenager logs a healthy snack.
  - The diabetic father logs a stable fasting glucose reading.
  - The kitchen utilizes leftovers to reduce food waste.
- This shifts the psychology from "individual restriction" to "collective familial success."

10. ENTERPRISE SCALABILITY & JURY HOOKS

10.1 Google Cloud: Architecture fundamentally relies on Vertex AI Search for Healthcare to securely ground LLM outputs in clinical truth, eliminating hallucination risks. **10.2 Swiggy/Quick Commerce:** Vernacular voice capabilities (Sarvam AI) combined with JIT (Just-in-Time) grocery routing capture the "Next Billion Users" in Tier 2/3 cities. **10.3 Quantiphi:** Implements rigorous Responsible AI (RAI) via hardcoded algorithmic paternalism that protects minors from toxic diet culture. **10.4 Vedanta (ESG Focus):** The exact algorithm optimizing diets for urban households is architecturally decoupled so it can be deployed on low-cost tablets across Vedanta's **Project Nand Ghar** (Anganwadis), digitizing rural nutrition and targeting childhood stunting at scale.